

CBSE 2027 Maths — Predicted Paper (Set 6 of 10)

Based on the official CBSE blueprint, recent competency-based modifications, and the most heavily repeated questions from previous year papers (PYQs), here is Set 6 of the predicted Class 12 Maths board paper. This set introduces all-new, high-probability problems across calculus, vectors, 3D geometry, and applied mathematics.

SECTION A: Objective Type Questions (1 Mark Each x 20 = 20 Marks)

- Q1.** Let $A = \{1, 2, 3, 4\}$. How many bijective (one-one and onto) functions can be defined from set A to itself? * **Chapter:** Relations and Functions * **Probability Tag: High**
- Q2.** Find the principal value of $\cot^{-1}(-\sqrt{3})$. * **Chapter:** Inverse Trigonometric Functions * **Probability Tag: Very High**
- Q3.** If A is a square matrix of order 3 and its determinant $|A| = -2$, find the value of $|-3A|$. * **Chapter:** Matrices and Determinants * **Probability Tag: Very High**
- Q4.** Using determinants, find the area of the triangle whose vertices are $(3, 8)$, $(-4, 2)$ and $(5, -1)$. * **Chapter:** Determinants * **Probability Tag: Medium**
- Q5.** What is the derivative of $2^{\sin x}$ with respect to x ? * **Chapter:** Continuity and Differentiability * **Probability Tag: High**
- Q6.** The volume of a sphere is changing. What is the rate of change of its volume with respect to its radius r when $r = 5$ cm? * **Chapter:** Application of Derivatives * **Probability Tag: High**
- Q7.** Evaluate the definite integral: $\int_0^{\pi/2} \cos^2 x \, dx$. * **Chapter:** Integrals * **Probability Tag: Very High**
- Q8.** Evaluate the indefinite integral: $\int \frac{1}{\sqrt{9-x^2}} \, dx$. * **Chapter:** Integrals * **Probability Tag: High**
- Q9.** Determine the degree of the differential equation: $x^2 \frac{d^2y}{dx^2} = \left(y + \frac{dy}{dx}\right)^{1/4}$. * **Chapter:** Differential Equations * **Probability Tag: Very High**
- Q10.** If $\vec{a}$ is a non-zero vector of magnitude a and $\lambda\vec{a}$ is a unit vector, find the value of the scalar λ (where $\lambda > 0$). * **Chapter:** Vector Algebra * **Probability Tag: Medium**
- Q11.** Find the angle between the vectors $\vec{p} = \hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{q} = 3\hat{i} - 2\hat{j} + \hat{k}$. * **Chapter:** Vector Algebra * **Probability Tag: High**
- Q12.** Write the vector equation of the line passing through the point $(1, -2, 3)$ and parallel to the vector $3\hat{i} - 2\hat{j} + 6\hat{k}$. * **Chapter:** Three-Dimensional Geometry * **Probability Tag: Very High**
- Q13.** What is the perpendicular distance of the point $P(3, -1, 11)$ from the y -axis? * **Chapter:** Three-Dimensional Geometry * **Probability Tag: High**

Q14. In a Linear Programming Problem, the objective function is $Z = 5x + 7y$. If the corner points of the bounded feasible region are $(0, 0)$, $(0, 4)$, $(3, 4)$, and $(4, 0)$, what is the maximum value of Z ? * **Chapter:** Linear Programming * **Probability Tag: Very High**

Q15. A fair die is thrown. Let A be the event "the number appearing is a multiple of 3" and B be the event "the number appearing is even". Find $P(A \cap B)$. * **Chapter:** Probability * **Probability Tag: High**

Q16. Evaluate: $\int \frac{e^x(1+x)}{\cos^2(xe^x)} dx$. * **Chapter:** Integrals * **Probability Tag: Very High**

Q17. What is the general solution of the differential equation $\frac{dy}{dx} = e^{x+y}$? * **Chapter:** Differential Equations * **Probability Tag: High**

Q18. If $P(A) = \frac{1}{2}$, $P(B) = \frac{3}{8}$, and $P(A \cup B) = \frac{5}{8}$, evaluate the conditional probability $P(A|B)$. * **Chapter:** Probability * **Probability Tag: Very High**

Q19. (Assertion-Reason): Assertion (A): The function $f(x) = |x - 1|$ is continuous at $x = 1$ but not differentiable at $x = 1$. **Reason (R):** The left-hand derivative and right-hand derivative of $f(x)$ at $x = 1$ exist but are not equal to each other. * **Chapter:** Continuity and Differentiability * **Probability Tag: High**

Q20. (Assertion-Reason): Assertion (A): The matrix $M = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$ is a symmetric matrix.

Reason (R): For any square matrix A with real number entries, $A + A'$ is always a symmetric matrix. * **Chapter:** Matrices * **Probability Tag: Medium**

SECTION B: Very Short Answer Type Questions (2 Marks Each x 5 = 10 Marks)

Q21. Prove that $\sin^{-1}\left(\frac{8}{17}\right) + \sin^{-1}\left(\frac{3}{5}\right) = \sin^{-1}\left(\frac{77}{85}\right)$. * **Chapter:** Inverse Trigonometric Functions * **Probability Tag: Very High**

Q22. If $y = \log(x + \sqrt{x^2 + a^2})$, prove that $\frac{dy}{dx} = \frac{1}{\sqrt{x^2 + a^2}}$. * **Chapter:** Continuity and Differentiability * **Probability Tag: High**

Q23. If $\vec{a}, \vec{b}, \vec{c}$ are three vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$ and $|\vec{a}| = 3, |\vec{b}| = 4, |\vec{c}| = 5$, find the exact value of $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$. * **Chapter:** Vector Algebra * **Probability Tag: Very High**

Q24. Find the particular solution of the differential equation $\frac{dy}{dx} = -4xy^2$, given that $y = 1$ when $x = 0$. * **Chapter:** Differential Equations * **Probability Tag: Medium**

Q25. A pair of dice is thrown 4 times. If getting a doublet is considered a "success," find the probability of getting exactly 2 successes using the binomial probability distribution. * **Chapter:** Probability * **Probability Tag: High**

SECTION C: Short Answer Type Questions (3 Marks Each x 6 = 18 Marks)

Q26. Check whether the relation R defined on the set $A = \{1, 2, 3, 4, 5, 6\}$ as $R = \{(a, b) : b = a + 1\}$ is reflexive, symmetric, or transitive. Justify your answer. *

Chapter: Relations and Functions * **Probability Tag: High**

Q27. If $A = \begin{pmatrix} 3 & -2 \\ 4 & -2 \end{pmatrix}$ and $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, find the scalar k such that $A^2 = kA - 2I$. *

Chapter: Matrices * **Probability Tag: Very High**

Q28. Differentiate $\tan^{-1} \left(\frac{\sqrt{1+x^2}-1}{x} \right)$ with respect to $\tan^{-1} x$. * **Chapter:** Continuity and Differentiability * **Probability Tag: Very High**

Q29. Evaluate the indefinite integral involving a linear numerator and quadratic denominator: $\int \frac{x+2}{2x^2+6x+5} dx$. * **Chapter:** Integrals * **Probability Tag: High**

Q30. Solve the linear differential equation: $\frac{dy}{dx} - 3y \cot x = \sin 2x$, given that $y = 2$ when $x = \frac{\pi}{2}$. * **Chapter:** Differential Equations * **Probability Tag: Very High**

Q31. A bag contains 4 white and 5 black balls. Another bag contains 3 white and 4 black balls. A ball is transferred blindly from the first bag to the second bag, and then a ball is drawn from the second bag. Find the probability that the ball drawn from the second bag is white. *

Chapter: Probability * **Probability Tag: Very High**

SECTION D: Long Answer Type Questions (5 Marks Each x 4 = 20 Marks)

Q32. Evaluate the following definite integral using standard properties of limits:

$\int_0^{\pi/2} \frac{\cos^2 x}{\cos^2 x + 4 \sin^2 x} dx$ * **Chapter:** Integrals * **Probability Tag: Very High**

Q33. An open box with a square base is to be made out of a given quantity of cardboard of area c^2 square units. Show that the maximum volume of the box is $\frac{c^3}{6\sqrt{3}}$ cubic units. *

Chapter: Application of Derivatives * **Probability Tag: Very High**

Q34. Find the shortest distance between the lines in 3D space given by the equations:

$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$. [1] * **Chapter:** Three-Dimensional Geometry *

Probability Tag: Very High

Q35. Using integration, find the area of the region bounded by the curves $y = x^2 + 2$, the line $y = x$, and the vertical ordinates $x = 0$, and $x = 3$. * **Chapter:** Application of Integrals *

Probability Tag: High

SECTION E: Case-Study Based Questions (4 Marks Each x 3 = 12 Marks)

Q36. Case Study 1: Linear Programming (Agricultural Optimization) A cooperative society of farmers has 50 hectares of land to grow two crops X and Y. The profit from crops X and Y per hectare are estimated as ₹10,500 and ₹9,000 respectively. To control weeds, a liquid herbicide has to be used for crops X and Y at rates of 20 litres and 10 litres per hectare, respectively. Further, no more than 800 litres of herbicide should be used overall in order to

protect fish and wildlife in a pond which collects drainage from this land. Let x and y be the land allocated to crops X and Y respectively. (i) Formulate the mathematical constraints for land allocation and herbicide usage. (1 Mark) (ii) Write the objective function Z to maximize the total profit. (1 Mark) (iii) Evaluate the corner points of the feasible region and determine how much land should be allocated to each crop to maximize the total profit. (2 Marks) *

Chapter: Linear Programming * **Probability Tag: Very High**

Q37. Case Study 2: Matrices (Market Revenue Analysis) A manufacturer produces three products x , y , and z which he sells in two distinct markets. Annual sales are indicated below: *Market I:* 10,000 units of x , 2,000 units of y , 18,000 units of z *Market II:* 6,000 units of x , 20,000 units of y , 8,000 units of z The unit sale prices of x , y , and z are ₹2.50, ₹1.50, and ₹1.00 respectively. (i) Express the sales volumes of both markets as a 2×3 matrix and the unit sale prices as a column matrix. (1 Mark) (ii) Find the total revenue collected in each market by using matrix multiplication. (1 Mark) (iii) If the unit production costs of the three commodities are ₹2.00, ₹1.00, and ₹0.50 respectively, formulate a cost matrix and find the total gross profit in both markets combined. (2 Marks) * **Chapter:** Matrices * **Probability Tag: High**

Q38. Case Study 3: Probability (Bayes' Theorem on Machine Defect Rates) In a corporate office, there are two primary photocopiers, Machine A and Machine B. Machine A is faster and produces 60% of the daily photocopies, while machine B produces the remaining 40%. Historically, it is known that 2% of the copies produced by Machine A are defective (too light, jammed, smeared), and 3% of the copies produced by Machine B are defective. At the end of the day, a single photocopy is selected at random from the output bin. (i) What is the probability that the selected photocopy is defective AND was produced by Machine A? (1 Mark) (ii) Use the Law of Total Probability to find the overall probability that the randomly selected photocopy is defective. (1 Mark) (iii) If the randomly selected photocopy is discovered to be defective, what is the exact probability that it was produced by Machine B? (2 Marks) *

Chapter: Probability * **Probability Tag: Very High**

CBSE Class 12 (2027) — AI-Predicted via PYQ/Blueprint Analysis. Probability-weighted guidance, not actual exam questions.